

Bootstrap — Topic 4: Typography ★ ★ ★

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end.

1. Typography Utilities kya hain?

Bootstrap typography utilities text ko **bina custom CSS likhe** style karne dete hain. Zyadatar text-related styling sirf classes add kar ke ho jati hai.

2. Headings

Normal HTML headings Bootstrap support karta hai:

```
html

<h1>Heading 1</h1>
<h2>Heading 2</h2>
...
<h6>Heading 6</h6>
```

Bootstrap Heading Classes

Kisi bhi element ko heading jaisa dikhaya ja sakta hai:

```
html

<p class="h1">Heading 1</p>
<p class="h2">Heading 2</p>
<p class="h3">Heading 3</p>
```

Ye `<p>` tags hain lekin heading jaisa appear karte hain — useful jab styling chahiye ho lekin HTML heading semantics nahi chahiye.

3. Display Headings

Ye normal headings se kaafi bare hote hain.

```
display-1
display-2
display-3
display-4
display-5
display-6
```

```
html

<h1 class="display-1">Display 1</h1>
<h1 class="display-6">Display 6</h1>
```

Class	Size
display-1	Huge
display-2	Very Large
display-3	Large
display-4	Medium Large
display-5	Medium
display-6	Smallest Display

Use case: Hero sections, landing pages, website banners.

```
html
```

```
<section class="text-center py-5">
  <h1 class="display-2">Welcome to My Website</h1>
  <p class="lead">Learn Bootstrap Easily</p>
</section>
```

4. Lead Text

`lead` paragraphs ko bara aur lighter bana deta hai.

```
html

<p class="lead">Hello World</p>
```

Used for: Introductions, Hero subtitles, Article summaries.

5. Text Alignment

```
text-start
text-center
text-end
```

```
html

<p class="text-start">Left</p>
<p class="text-center">Center</p>
<p class="text-end">Right</p>
```

Responsive Alignment

Screen size ke hisab se alignment change kar sakte hain:

```
html
```

```
<p class="text-center text-md-start">  
    Center on mobile, Left on tablet and larger.  
</p>
```

Small screens → CENTER, Medium+ → LEFT.

6. Font Weight

```
fw-bold  
fw-bolder  
fw-semibold  
fw-medium  
fw-normal  
fw-light  
fw-lighter
```

```
html  
  
<p class="fw-bold">Bold</p>  
<p class="fw-light">Light</p>  
<p class="fw-normal">Normal</p>
```

7. Font Style

```
fst-normal  
fst-italic
```

```
html
```

```
<p class="fst-italic">This is italic.</p>
<p class="fst-normal">Normal text.</p>
```

8. Text Transform

Original content change kiye bina text ko transform karta hai.

```
text-uppercase
text-lowercase
text-capitalize
```

html

```
<p class="text-uppercase">hello world</p>    → HELLO WORLD
<p class="text-lowercase">HELLO WORLD</p>    → hello world
<p class="text-capitalize">hello world from bootstrap</p>    → Hello World From
```

9. Text Wrapping

Kabhi kabhi lamba text overflow ho jata hai.

```
text-wrap
text-nowrap
```

html

```
<div class="w-25 border">
  <span class="text-nowrap">ThisIsAVeryLongWordThatWillNotBreak</span>
</div>
```

Text ek line par rehta hai aur overflow ho sakta hai.

```
html

<div class="w-25 border">
  <span class="text-wrap">Bootstrap allows wrapping.</span>
</div>
```

Ab text next line par wrap ho jata hai.

10. Text Decoration

```
text-decoration-none
text-decoration-underline
text-decoration-line-through
```

```
html

<a href="#" class="text-decoration-none">No Underline</a>
<p class="text-decoration-underline">Underlined</p>
<p class="text-decoration-line-through">Old Price</p>
```

Useful for: Discount prices, Links, Deleted items.

11. Text Colors

```
text-primary    text-secondary    text-success
text-danger     text-warning     text-info
text-light      text-dark         text-body
text-muted      text-white
```

```
html
```

```
<p class="text-primary">Primary</p>
<p class="text-success">Success</p>
<p class="text-danger">Danger</p>
```

Common meanings

Class	Color	Typical Use
text-primary	Blue	Main branding
text-success	Green	Success messages
text-danger	Red	Errors
text-warning	Yellow	Warnings
text-info	Cyan	Information
text-secondary	Gray	Secondary text
text-muted	Light Gray	Less emphasis
text-dark	Black/Dark	Main content
text-light	Light Gray	Dark backgrounds
text-white	White	Dark backgrounds

12. Multiple Utilities Together

Bootstrap classes ek sath combine hone ke liye design hui hain.

```
html
```

```
<h1 class="display-3 fw-bold text-primary text-center">
  Bootstrap Typography
</h1>

<p class="lead text-center fst-italic text-secondary">
  Learn Bootstrap the Easy Way
</p>
```

Yahan `display-3` (large heading) + `fw-bold` (bold) + `text-primary` (blue) + `text-center` (center align) + `lead` (larger paragraph) + `fst-italic` (italic) + `text-secondary` (gray) — sab ek sath apply ho rahi hain.

13. Mini Project: Hero Section

```
html

<div class="container py-5 text-center">
  <h1 class="display-2 fw-bold text-primary">
    Learn Bootstrap
  </h1>
  <p class="lead text-secondary">
    Build responsive websites faster than ever.
  </p>
  <p class="fw-light">
    No custom CSS needed for basic typography.
  </p>
</div>
```

Isme use hui classes: `display-2`, `fw-bold`, `lead`, `text-primary`, `text-secondary`, `fw-light`, `text-center`, `container`, `py-5`.

14. Summary Table

Topic	Classes
Normal headings	<code><h1></code> – <code><h6></code> , <code>.h1</code> – <code>.h6</code>
Display headings	<code>display-1</code> to <code>display-6</code>
Lead text	<code>lead</code>
Text alignment	<code>text-start</code> , <code>text-center</code> , <code>text-end</code>
Responsive alignment	<code>text-md-center</code> , <code>text-lg-end</code> , etc.
Font weight	<code>fw-bold</code> , <code>fw-light</code> , <code>fw-normal</code> , <code>fw-semibold</code> , etc.
Font style	<code>fst-italic</code> , <code>fst-normal</code>
Text transform	<code>text-uppercase</code> , <code>text-lowercase</code> , <code>text-capitalize</code>
Text wrapping	<code>text-wrap</code> , <code>text-nowrap</code>
Text decoration	<code>text-decoration-none</code> , <code>text-decoration-underline</code> , <code>text-decoration-line-through</code>
Text colors	<code>text-primary</code> , <code>text-success</code> , <code>text-danger</code> , <code>text-warning</code> , <code>text-info</code> , <code>text-muted</code> , <code>text-dark</code> , <code>text-light</code> , <code>text-white</code>

Practice Exercises

1. Ek hero section banao jisme `display-1` heading aur `lead` paragraph ho, dono center-aligned.
 2. Ek pricing card banao jahan old price `text-decoration-line-through` use kare aur discounted price `text-success` + `fw-bold` use kare.
 3. Ek paragraph banao jo mobile par centered ho aur medium screens aur upar left-aligned ho (responsive alignment classes se).
 4. Teen paragraphs display karo `text-uppercase`, `text-lowercase`, aur `text-capitalize` ke sath, farq observe karo.
 5. `text-success`, `text-warning`, `text-danger`, aur `text-info` use kar ke status messages ki list banao.
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Active Recall — Bootstrap Typography

┆ Khud se answer karne ki koshish karo, phir neeche scroll karke check karo.

Q1. `<h1>` tag aur `.h1` class mein kya farq hai? `.h1` class kab useful hoti hai?

Q2. Display headings normal headings se kaise different hain, aur ye typically kahan use hoti hain?

Q3. `lead` class kya karti hai, aur ye typically kis cheez ke liye use hoti hai?

Q4. Responsive text alignment ka example do — `text-center text-md-start` ka behavior mobile aur desktop par kya hoga?

Q5. Font weight ke liye konsi classes available hain (koi 4 likho)?

Q6. Text transform ki teen classes kya hain, aur `text-capitalize` "hello world from bootstrap" ko kaise transform karegi?

Q7. `text-wrap` aur `text-nowrap` mein farq kya hai?

Q8. Text decoration ki teen classes kya hain, aur `text-decoration-line-through` typically kahan use hoti hai?

Q9. `text-success`, `text-danger`, aur `text-muted` ka typical use case kya hai?

Q10. Ek heading banao jo large ho, bold ho, blue color mein ho, aur center-aligned ho — konsi classes combine karogay?

Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

A1. `<h1>` ek actual HTML heading tag hai jo semantic meaning bhi carry karta hai (SEO, accessibility ke liye important). `.h1` class kisi bhi element (jaise `<p>`) ko heading jaisa **look** de deti hai bina us element ko actual heading semantics dene ke — useful jab visually heading chahiye lekin HTML structure heading na ho.

A2. Display headings (`display-1` se `display-6`) normal headings se kaafi bari hoti hain — `display-1` sabse badi, `display-6` sabse choti (par phir bhi normal headings se badi). In ka use hero sections, landing pages, aur website banners mein hota hai.

A3. `lead` class paragraph ko larger aur lighter bana deti hai. Ye introductions, hero subtitles, aur article summaries ke liye use hoti hai.

A4. `text-center` `text-md-start` mein small screens (mobile) par text center mein hoga, aur medium screens ($\geq 768\text{px}$) aur uske upar text left-aligned ho jayega.

A5. Koi bhi 4: `fw-bold`, `fw-bolder`, `fw-semibold`, `fw-medium`, `fw-normal`, `fw-light`, `fw-lighter`.

A6. Teen classes: `text-uppercase`, `text-lowercase`, `text-capitalize`. `text-capitalize` "hello world from bootstrap" ko "Hello World From Bootstrap" mein transform karegi (har word ka first letter capital).

A7. `text-nowrap` text ko ek hi line par force karta hai (overflow ho sakta hai), jabke `text-wrap` text ko normal tarah next line par wrap hone deta hai.

A8. Teen classes: `text-decoration-none`, `text-decoration-underline`, `text-decoration-line-through`, `text-decoration-line-through` typically discount prices (old price strike-through) ya deleted items ke liye use hoti hai.

A9. `text-success` (green) success messages ke liye, `text-danger` (red) errors ke liye, aur `text-muted` (light gray) less-emphasis text ke liye use hoti hai.

A10. `display-3 fw-bold text-primary text-center` (ya similar combination) — `display-*` large size, `fw-bold` bold weight, `text-primary` blue color, aur `text-center` center alignment ke liye.

</details>